Blayne Griffin

bgriffin.astro@gmail.com

Education

University of Central Arkansas, **B.S. Physics** — *Conway, Arkansas*
Class of Dec. 2025

- Graduated with 3.40 cumulative GPA
- **Relevant Courses:** Observational Techniques, Electromagnetism, Computational Physics, Thermal Physics, Classical Mechanics, Quantum Physics, Calculus 1-3, Differential Equations

Lyon College, **B.S. Chemistry** — *Batesville, Arkansas*
Class of May 2024

- Honors Fellow, graduated with 3.315 cumulative GPA
- **Relevant Courses:** Thermodynamics, Quantum Chemistry, Organic Chemistry, Advanced Inorganic Chemistry, Linear Algebra, Modern Physics, Astrophotography, Speleology

Arkansas School for Mathematics, Sciences, and the Arts — *Hot Springs, Arkansas*
Class of May 2021

- Graduated with 3.946 weighted GPA
- Arkansas residential collegiate high school
- Presented research over Carbon Nanoparticles, earning First in regional ISEF science fair and Third in the State.

Work Experience

Research and Outreach Assistant, Hat Creek, CA — *SETI Institute*
Jan. 2026 - Aug. 2026

- Monitored and operated telescope systems for more than 400 on-sky hours, trained new operators, and developed telescope control interfaces in Python. Interfaces have been used for 6+ projects since launch in 2026.
- Performed observation and analysis for a first-author SETI paper on the “One of everything” Exotica catalog
- Gave tours to 1,000+ guests, educating them on radio astronomy fundamentals
- Provided staff support for both the SETI Institute and Breakthrough Listen REUs and led education workshops for students

Technosignature Research Intern, Mountain View, CA — *SETI Institute*
June 2025 - Aug. 2025

- Began novel technosignature search of Breakthrough Listen “One of Everything” 816 target catalog, culminating in REU Award of Excellence
- Wrote observing script to observe the comet 3I/ATLAS

within hours of discovery

Part-time Educator — *Museum of Discovery*, *May 2023 - Dec. 2025*

- Provided creative direction and developed educational material for Tesla Coil show
- Facilitated guest experience and education regarding scientific topics, including electrodynamics, thermodynamics, optics, and acoustics.
- Received Culture Award for independent design, creation, and implementation of a tactile Tesla Coil model for the blind and visually impaired

Physics Tutor — *Lyon College*, *Aug. 2022 - May 2023*

- Tutored peers in Physics and related Mathematics

Selected Peer-Reviewed Research Projects and Publications

- **Blayne Griffin**, Sofia Sheikh (PI) et al. "One of Everything: A technosignature search of the Exotica Catalog with the Allen Telescope Array"
  - Performed a novel observation of an 816 target catalog, spanning solar system to extragalactic targets. Searched this catalog for technosignatures of "life as we don't know it", setting upper limits for the ~650 targets observable from our coordinates.
  - Created a novel python interface to control the Allen Telescope Array, which has been used for 5+ projects since launch in 2026
  - Presented as invited research talk at Penn State SETI Symposium
  - Presented at the National Conference on Undergraduate Research
  - Presented at the American Astronomical Society's 247th meeting as a Chambliss Finalist, earning an Honorable mention
    - https://aas247-aas.ipostersessions.com/default.aspx?s=6D-A3-85-92-09-9B-88-BD-A7-E8-50-A0-21-DC-2C-C3

- Sofia Sheikh (PI) et al. including **Blayne Griffin**, "A Search for Radio Technosignatures from Interstellar Object 3I/Atlas with the Allen Telescope Array"
  - Worked with a team to develop an observing script of a novel interstellar object mere hours after its official announcement and search for technosignatures from 1-11 GHz
  - **Published** in the Astronomical Journal
    - https://iopscience.iop.org/article/10.3847/1538-3881/ae6651

- August Childress, **Blayne Griffin**, Jeremy Lusk (PI), "Radio Observations of the April 8th, 2024 Total Solar Eclipse"
  - Developed a python pipeline to process collected single dish data and calculate the doppler shift of observed data.
  - Presented:
    - Chambliss Finalist at American Astronomical Society 245

        - https://aas245-aas.ipostersessions.com/?s=DB-CA-C1-ED-B0-6A-40-45-1D-AE-69-F4-7B-D7-AF-BB
    - STEM Posters at the Arkansas State Capitol
    - University of Central Arkansas College of Science and Engineering Symposium
- October 1, 2025 **publication** in Volume 34 of the Journal for Undergraduate Research in Physics and Astronomy (JURPA) https://students.aip.org/jurpa/radio-telescope-observations-of-the-april-8-2024-total-solar-eclipse-at-1420-mhz

- Carl B. Hollandsworth (PI), **Blayne M. Griffin** et al. "Co-crystallization of isomers of iron(ii) complex of a tridentate mixed amine/imine ligand in one unit cell", Mendeleev Communications, Volume 34, Issue 1, 2024, Pages 47-50, ISSN 0959-9436
    - Used X-Ray Crystallography to analyze novel inorganic iron complexes
        - https://doi.org/10.1016/j.mencom.2024.01.014

- Carl B. Hollandsworth (PI), **Blayne M. Griffin** et al. "Synthesis and characterization of iron(II) Schiff-base complexes of tridentate mixed amine/imine ligands with cis- and trans-1,2-diaminocyclohexane backbones", Polyhedron, Volume 246, 2023, 116669, ISSN 0277-5387
    - Utilized self-assembly to synthesize a novel inorganic iron complex, further analyzing with FT-IT, UV-VIS, NMR, and TLC
        - https://doi.org/10.1016/j.poly.2023.116669

Honors and Awards

Chambliss Award Honorable Mention *Jan. 2026*

SETI Institute REU Award of Excellence *Aug. 2025*

Chambliss Award Finalist at American Astronomical Society 245 *Jan. 2025*

Above and Beyond Award *April 2024*
Presented to one Lyon College student per year who goes above and beyond the status quo to improve life and experience for Lyon College and its students.

Lyon College Honors Fellows Program *2021-2024*
Graduated on an accelerated Honors track. Requires a 3.25 Cumulative GPA, advanced classes with an interdisciplinary focus, and 2 independent projects that go above and beyond standard coursework.

Physics Book Award *April 2024*
Presented to one Physics student per year doing excellent independent work in Physics.

Lyon College Mortar Board *2023-2024*
A national honor society devoted to academic excellence and community involvement

James Alexander Shanks Chemistry Award *April 2023*
Presented to one Chemistry student per year doing excellent independent work in Chemistry

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Instrumentation and Skills

Observational Astronomy:

- Allen Telescope Array
    - 400+ hours of active observing experience with correlation and beamforming on an interferometric array.
    - Developed telescope control interface in Python
    - Trained new observers and led educational workshops for undergraduates
- Spider 230C Radio Telescope
    - Approximately 48 hours of active observing, including the Crab Pulsar, Cassiopeia A, and the Sun
    - Developed post-processing pipeline to determine the Doppler Shift of observed targets
- Meade-14 Optical Telescope
    - Approximately 48 hours of active observing, including CCD calibration, asteroid astrometry, Monochrome and LRGB post-processing, and stellar photometry
- Celestron C8 Optical Telescope
    - Approximately 200+ hours of active observing, including Jupiter, Saturn, the Moon, and solar prominences during a Total Solar Eclipse

Software:

- Programming
    - Python (Advanced)
        - Pandas, Bash, Astropy, Numpy, Scipy, Matplotlib, TKinter, CustomTKinter
    - Linux (Intermediate)
    - R (Beginner)
    - Java (Beginner)
    - HTML (Beginner)
- Astronomy
    - Siril
    - Astrometrica
    - AstroImageJ
    - PIPP
    - GIMP
    - Autostakkert!
    - BLISS - Radio technosignature search pipeline
    - NBeamAnalysis - Radio narrowband search pipeline

Instrumental Analysis:

- Spectroscopy
    - Fluorescence Spectrophotometer
    - Ultraviolet-Visible Spectroscopy
    - Infrared Spectroscopy
    - Nuclear Magnetic Resonance Spectroscopy
    - Gas Chromatography
    - High Performance Liquid Chromatography
    - Thin Layer Chromatography

Chemical Synthesis:

- Organic and Inorganic synthesis
- Crystallization
- Sample Plating

Professional Membership and Leadership

Sigma Pi Sigma *April 2026 - Present*

- Member of Physics Honor Society dedicated to excellent scholarship in Physics and Astronomy

Society of Physics Students *Aug. 2024 - Dec. 2025*

- Officer At Large
- Alumni member

Institute of Electronics and Electrical Engineers (IEEE) *Dec. 2024 - Dec. 2025*

- Alumni member

Solar Eclipse Committee *Aug. 2023-April 2024*

- Only student chosen to join the committee responsible for planning events regarding the April 8, 2024, Solar Eclipse.
- Organized and presented an Astrophotography Workshop focused on teaching guests how to photograph the solar eclipse with a cell phone.
- Earned “Above and Beyond” Award for service

Zeta Beta Tau Fraternity, Phi Theta Gamma Chapter *April 2022 –Present*

- Vice President
- Operations Director
- Brotherhood Development Director
- Risk Prevention and Wellness Director
- Alumni Committee Chair

Inter-Fraternity Council *Jan. 2023-May 2024*

- Vice President of Records
- Chapter Representative

Mortar Board *April 2023-May 2024*
- Member of a National Honor Society dedicated to academic excellence and community involvement

Chemistry Candidate Search *Spring 2024*
- One of two students chosen to assist faculty in hiring a new tenure track Chemistry professor

American Chemical Society *2024*
- Student member

Outreach and Service

- Invited Speaker at Mt. Shasta Audubon Society event for Perseids Meteor Shower *Summer 2026*
- Invited judge at Mt. Shasta High School theoretical evolution AP Class Fair *Spring 2026*
- Hosted educational workshops for Society of Physics Students *2025*
- Designed and implemented tactile Tesla Coil model for the blind and visually impaired at the Museum of Discovery *Summer 2024*
- Hosted astrophotography workshop on safe cell phone astrophotography of the April 2024 Total Solar Eclipse *Spring 2024*
- Used a student grant to obtain a dedicated astrophotography camera I then used to host an eclipse livestream to 1,000+ attendees *2024*
- Invited co-host for public museum outreach at the Park Plaza Mall *2024*
- Organized and held “Inferno” charity event, raising $1,000+ for Children’s Miracle Network *2021-2024*
- Volunteered at Batesville Animal shelter to clean and take care of animals *2021*
- Volunteered to clean Polk Bayou *2021-2024*

Other Research

- Ben Jacobson-Bell et al. including **Blayne Griffin**, “Nondetection of Narrowband Radio Technosignatures from LHS 1140 b at 1–12 GHz”
    - Published in the Research Notes of the AAS
    - https://doi.org/10.3847/2515-5172/ae91e1

References available on request